Abdullah Sheikh

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Education

University of Calgary — Calgary, AB
Bachelor of Computer Science — *Expected June 2028*

Technical Skills

Languages: C, C++, Python, Java, Assembly (ARM), Ruby, SQL, Fortran
ML Systems: Supervised/Unsupervised Learning, Reinforcement Learning, CNN, GBDT
Parallel & GPU Computing: CUDA, Metal, OpenMP, ROCm, GPU/SoC Micro-architecture
Systems & Infrastructure: QNX RTOS, Linux, Embedded Drivers, Interrupt-driven I/O, AWS

Experience

Dominion Dynamics — Jan. 2027 – Apr. 2027
Software Engineering Intern — *Ottawa, ON*
- Joining in Winter 2027 to build autonomous systems for Arctic defence.

AMD — Apr. 2026 – Present
Software Engineering Intern — *Calgary, AB*
- Wrote C++ HIP compute kernels for the ROCm stack on Instinct data centre GPUs for HPC/ML applications.
- Improved rocFFT kernel execution times by 5x by using fused Bluestein kernels for prime length FFTs.
- Built a Fortran generator to interface with ROCm headers, adding bindings for 6+ internal ROCm libraries.
- Added SplitMix64 verification to validate bitwise consistency for GPU FFTs, outperforming SHA-256 by 186x.

Shopify — Jan. 2026 – Apr. 2026
Software Engineering Intern — *Toronto, ON*
- Cut circuit breaker CPU ∼20% by profiling and rewriting a hot path in a performance-critical C extension.
- Shipped a fault-injection integration letting 100+ teams validate system behaviour under simulated failures.
- Root-caused incidents across 5+ services, documenting findings in a monitoring tool adopted org-wide.

BlackBerry — Sep. 2025 – Dec. 2025
Software Engineering Intern (QNX) — *Ottawa, ON*
- Shipped 5+ C/C++ drivers on the QNX RTOS, verifying behaviour against hardware functional specifications.
- Designed interrupt-driven I/O on ARM SoCs, hitting sub-millisecond latency across 8+ hardware peripherals.
- Delivered 3x throughput on a C++ pipeline by fixing cache-unfriendly memory access patterns found in GDB.

Enverus — May 2025 – Aug. 2025
Software Engineer Intern — *Calgary, AB*
- Cut $200K+ in annual costs by designing AWS PrivateLink network infrastructure across 350+ VPCs.
- Built 15+ isolated AWS sandboxes, documenting workflows so teams could safely validate infra changes.

Projects

RLHF with GRPO on Qwen2.5-1.5B | *Python, PyTorch, TRL, vLLM* — Nov. 2026
- Trained two HH-RLHF reward models (74% accuracy); fine-tuned Qwen2.5-1.5B via GRPO on 8 NVIDIA H100s.
- Raised red-team refusals from 62.5% to 94% with a combined reward; caught harmless-only reward hacking.

PPO Reinforcement Learning Agents | *Python, PyTorch, ROCm, Unity ML-Agents* — Nov. 2026
- Trained PPO agents in Unity ML-Agents on NVIDIA H100 GPUs, lifting SnowballTarget reward 6×.
- Scaled MA-POCA multi-agent self-play to 10M steps for 2v2 soccer, raising ELO from 1167 to 1660.
- Ported CleanRL's PyTorch PPO to Gymnasium, training at 2,300 steps/s on an NVIDIA Blackwell GPU.

Leadership Experience

BobaTalks Mentor — Feb. 2026 – Present
BobaTalks — *Toronto, ON*
- Coaching early-career students through 1:1 sessions on technical growth and industry navigation.

Certifications & Activities

Hackathons: Hack the North @ UWaterloo (2025 & 2026), Hack the Change @ UCalgary (2025)
Open Source Contributions: QNX Everywhere, Semian, ROCm
AWS: Cloud Practitioner Essentials, Bedrock Workshop, Builder's Quest Badge